

Kavya Infoweb Pvt. Ltd. (KIPL)

HRMS Portal - Full Requirement Sheet

React Frontend | Django Backend | PostgreSQL Database

Document Type	Project Requirement Sheet
Project Name	HRMS Portal
Project Type	Web Application
Target Users	Super Admin, HR, Project Manager, Team Lead, Employee
Prepared For	Internal development team and new employee onboarding
Prepared Date	05 May 2026
Version	Final Full Version 1.0

Purpose

This document defines the complete HRMS Portal scope, role-wise work, access permissions, modules, submodules, pages, data models, frontend requirements, backend requirements, workflow, and remaining development plan. It is prepared to guide the frontend, backend, database, QA, and deployment teams without confusion.

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1. Executive Summary

- The HRMS Portal is a centralized application for managing company employees, HR operations, attendance, leave, payroll, announcements, assets, project teams, and technical support tickets.
- The system must work with strict role-based access control so every user sees only the pages and actions allowed for their role.
- The uploaded project currently contains a React + Vite frontend with protected routes, role dashboards, sidebar menus, dummy data, and static pages. Backend APIs and real database integration are not included in the uploaded React project.
- The final implementation should connect React with Django REST APIs and PostgreSQL so login, employee, attendance, leave, payroll, assets, announcements, support tickets, and settings data are stored in the database.

Current Uploaded Project Findings

Area	Finding
Frontend framework	React 18 with Vite, React Router DOM, Bootstrap, Remixicon.
Routing	HashRouter is used. Login route and role-protected route groups are present.
Roles found	admin, hr, teamLead, projectManager, employee.
Existing pages found	Login, Admin Dashboard, HR Dashboard, Project Manager Dashboard, Team Lead Dashboard, Employee Dashboard, Employees, User Management, Attendance, Leave Requests, Payroll, Announcements, Tasks, Projects, My Team, Support Tickets, Profile, Settings.
Data source	Static dummyData.js is used for people, attendance, leave, announcements, tasks, projects, salary records, and dashboard stats.
Backend status	No Django backend found inside this React zip. Backend needs to be created or connected separately.
Database status	No PostgreSQL connection found in React project. Database integration is pending.
Assets module	Asset management is required for HRMS but not found as a complete page/module in the uploaded React project. It should be added.

2. Project Scope

In Scope

- Role-based login and dashboard redirection.
- Employee profile and employee management.
- Attendance tracking, attendance reports, check-in/check-out, and monthly summaries.
- Leave request, leave approval, leave history, and leave balance management.
- Payroll and salary/payslip management.
- Project, team, and task management for PM and TL roles.
- Announcement management where only Super Admin and HR can create announcements; all other roles can view them.
- Support ticket system with title, description, category, priority, status, and screenshot upload.
- Asset management for company laptops, desktops, mobile phones, monitors, keyboards, mice, headphones, tablets, and other assets.
- Settings, profile, password change, and basic system configuration.
- Django REST backend with PostgreSQL database and secure permissions.

Out of Scope for First Version

- Advanced biometric device integration.
- Complex multi-company HRMS setup.
- AI-based performance analytics.
- Advanced accounting integrations.

- Mobile app version. First version is web responsive only.

3. Technology Stack

Layer	Recommended Technology	Purpose
Frontend	React with JavaScript, Vite, React Router DOM, Bootstrap/CSS	Build responsive role-based user interface.
Backend	Django and Django REST Framework	Create APIs, authentication, permissions, business logic.
Database	PostgreSQL	Store all HRMS operational data.
Authentication	JWT or secure Django session authentication	Secure login and protected APIs.
Storage	Django media storage / cloud storage later	Store ticket screenshots, employee documents, asset images.
Deployment	Netlify for frontend, Render/Railway/VPS for backend, PostgreSQL cloud database	Host application live.

4. User Roles and Responsibilities

Role	Main Responsibility
Super Admin	Complete owner of the HRMS system. Manages users, roles, employees, attendance, leave, payroll, announcements, support, assets, and settings.
HR	Handles HR operations such as employees, attendance review, leave approval, payroll processing, announcements, support tickets, and asset assignments.
Project Manager	Manages projects, project teams, project tasks, project attendance view, project leave visibility, and team performance.
Team Lead	Manages assigned team members, assigns tasks, reviews attendance, recommends leave requests, views announcements, and raises support tickets.
Employee	Uses self-service features such as profile, attendance, leave request, payslip view, assigned assets, announcements, and support tickets.

5. Complete Module List

No.	Module	Description
1	Authentication and Authorization	Login, logout, protected routes, role detection, dashboard redirect, permission rules.
2	Dashboard	Role-specific cards, counts, quick actions, recent activity, charts, reminders.
3	User Management	Create/manage user accounts, assign roles, activate/deactivate access.
4	Employee Management	Employee records, departments, designations, joining details, reporting manager.
5	Attendance Management	Daily attendance, check-in/out, monthly records, late/absent reports.
6	Leave Management	Apply leave, approve/reject, recommend, leave balance, leave history.
7	Payroll/Salary Management	Salary structure, monthly payroll, payslip generation, deductions, status.
8	Project Management	Projects, teams, milestones, task relation, progress status.
9	Task Management	Assign tasks, update status, priority, due date, owner.
10	My Team	View team members, reporting hierarchy, attendance and workload summary.
11	Asset Management	Manage assets, assign assets, replacements, asset return, repair status.
12	Announcement Management	Create and view company announcements. Create access only for Admin and HR.
13	Support Ticket Management	Raise issue, attach screenshot, track ticket, manage resolution.
14	Profile Management	Personal details, contact info, password update, profile photo.
15	Settings	Company settings, role permissions, departments, leave types, payroll settings.

6. Module and Submodule Definitions

Authentication and Authorization

Every user must login using valid credentials. After login, the system reads the role and redirects to the correct dashboard. Users must not access unauthorized routes by manually changing the URL.

Submodules / Pages
Login page
Logout action
Forgot password later
Role-based dashboard redirect
Protected routes
Invalid access handling

Dashboard

Each role dashboard shows relevant summary cards, quick actions, recent activity, notifications, and pending work.

Submodules / Pages
Admin Dashboard
HR Dashboard
Project Manager Dashboard
Team Lead Dashboard
Employee Dashboard

User Management

Super Admin manages all application accounts. HR may view employee account status if allowed, but only Super Admin should control role changes in first version.

Submodules / Pages
User list
Add user
Edit user
Role assignment
Activate/deactivate user
Reset password

Employee Management

Admin and HR maintain employee master data. PM and TL can view assigned team members only. Employee can view own profile only.

Submodules / Pages
Employee list
Add employee
Edit employee
Employee profile
Department/designation mapping
Reporting manager assignment

Attendance Management

Employees mark own attendance. HR/Admin can view all. PM/TL can view assigned team attendance only.

Submodules / Pages
Mark attendance
Check-in/check-out
Attendance table
Monthly attendance report
Late/absent status
Team attendance view

Leave Management

Employees, TL and PM can apply leave. TL and PM may recommend/team review. HR/Admin approve or reject final leave requests.

Submodules / Pages
Apply leave
Leave request list
Leave approval
Leave recommendation
Leave balance
Leave history

Payroll/Salary Management

HR/Admin manage payroll. Employees, TL and PM should only view their own payslip and salary details.

Submodules / Pages
Salary structure
Payroll list
Payslip view
Deductions
Bonus/allowance
Payroll status

Project Management

Project Manager manages projects and team assignments. Admin can view/control all projects.

Submodules / Pages
Project list
Create project
Assign team
Project progress
Milestones
Project status

Task Management

PM and TL assign tasks. Employees update assigned task status. Admin can view all.

Submodules / Pages
Task list
Create task

Submodules / Pages
Assign task
Update task status
Priority
Due date

Asset Management

HR/Admin manage company assets. Employees can view assigned assets and raise replacement/repair requests.

Submodules / Pages
Asset dashboard
Manage assets
Asset assignment
Replacement request
Repair status
Return asset

Announcement Management

Only Admin and HR can post announcements. PM, TL and Employee can view announcements only.

Submodules / Pages
Announcement list
Create announcement
Edit announcement
Delete announcement
Priority/category

Support Ticket Management

All roles can raise support tickets. Admin and HR manage and resolve tickets.

Submodules / Pages
Raise ticket
Ticket list
Ticket details
Screenshot upload
Status tracking
Admin/HR resolution

Profile Management

Every user can view and update limited personal information. Sensitive data updates require HR/Admin approval.

Submodules / Pages
View profile
Edit contact details
Change password
Profile photo

Settings

Admin manages full settings. HR may manage HR-related settings only if allowed.

Submodules / Pages
Company settings
Departments
Designations
Leave types
Role permissions
Payroll configuration

7. Full Role-wise Page Access Matrix

Legend: Full = create, view, update and delete where applicable. Manage = operational control. View = read only. Own = only own data. Team = assigned team only. No = no access.

Page / Feature	Super Admin	HR	Project Manager	Team Lead	Employee
Login	Full	Full	Full	Full	Full
Admin Dashboard	Full	No	No	No	No
HR Dashboard	No	Full	No	No	No
PM Dashboard	No	No	Full	No	No
TL Dashboard	No	No	No	Full	No
Employee Dashboard	No	No	No	No	Full
User Management	Full	View user status	No	No	No
Employee List	Full	Full	Team/View	Team/View	No
Add Employee	Full	Full	No	No	No
Edit Employee	Full	Full	No	No	Own limited profile only
Attendance Page	Full all	Full all	Team/View	Team/View	Own
Mark Attendance	No	Own if HR user	Own	Own	Own
Attendance Reports	Full all	Full all	Project team	Team	Own
Apply Leave	Own	Own	Own	Own	Own
Leave Approval	Final approve	Final approve	Recommend/Team review	Recommend/Team review	No
Leave History	All	All	Team	Team	Own
Payroll Management	Full	Full	No	No	No
Payslip View	All	All/Own	Own	Own	Own
Project List	Full	View	Manage assigned	View assigned	No
Create/Edit Project	Full	No	Full assigned	No	No
Task Assignment	Full	View	Manage project tasks	Manage team tasks	No
Task Status Update	View all	View	Manage	Manage	Own assigned tasks
My Team	All teams	All employees	Project team	Assigned team	No
Asset Dashboard	Full	Full	View team assets	View team assets	Own assigned assets
Manage Assets	Full	Full	No	No	No
Asset Assignment	Full	Full	No	No	No
Replacement Request	Manage all	Manage all	Raise/View team	Raise/View team	Raise own
Create Announcement	Full	Full	No	No	No
View Announcements	Full	Full	Full	Full	Full
Support Ticket Raise	Full	Full	Full	Full	Full
Support Ticket Manage	Full	Full	No	No	No
Profile	Own	Own	Own	Own	Own
Settings	Full	Limited HR settings	No	No	No

8. Role-wise Work Details

Super Admin

- Manage complete system access, roles and permissions.
- Create/update/delete employees and users.
- View all attendance, leave, payroll, projects, assets, announcements and support tickets.

- Control settings such as departments, designations, leave types and payroll configurations.
- Monitor overall HRMS data and company activity.

HR

- Add and maintain employee data.
- Review attendance and leave requests.
- Approve/reject employee leave requests.
- Prepare payroll and salary records.
- Assign and track company assets.
- Create company announcements.
- Manage and resolve support tickets.

Project Manager

- View PM dashboard and assigned projects.
- Create/update assigned projects and milestones.
- Manage project team and assigned tasks.
- View team attendance and leave status.
- Review project performance and pending work.
- Raise support tickets and view announcements.

Team Lead

- View TL dashboard and assigned team.
- Assign tasks to team members.
- Review team attendance.
- Recommend leave requests before HR/Admin approval.
- Track task progress and priority.
- Raise support tickets and view announcements.

Employee

- View employee dashboard.
- Mark own attendance.
- Apply for leave and view leave history.
- View salary/payslip.
- View assigned assets and raise replacement/repair request.
- View announcements and raise support tickets.
- Update own profile details where allowed.

9. Complete Page Inventory

Module	Page	Route	Role Access	Purpose
Authentication	Login	/login	All users	User login and role selection/verification.
Dashboard	Admin Dashboard	/admin/dashboard	Admin	Overall employee, attendance, leave, payroll and system summary.

Module	Page	Route	Role Access	Purpose
Dashboard	HR Dashboard	/hr/dashboard	HR	HR-focused summary with employees, interviews, leave approvals and announcements.
Dashboard	PM Dashboard	/project-manager/dashboard	PM	Project, milestone, review and capacity summary.
Dashboard	TL Dashboard	/team-lead/dashboard	TL	Team attendance, tasks and leave summary.
Dashboard	Employee Dashboard	/employee/dashboard	Employee	Personal attendance, leave, tasks and payslip summary.
User	User Management	/admin/users	Admin	Manage user accounts and role assignment.
Employee	Employees	/admin/employees, /hr/employees	Admin, HR	Employee list and employee master data.
Attendance	Attendance	role-wise /attendance	All roles	Attendance view based on role scope.
Leave	Leave Requests/Approval	role-wise leave routes	All roles	Apply, review, approve and view leave history.
Payroll	Payroll/Salary	role-wise /payroll	All roles limited	Admin/HR manage; others view own payslip.
Project	Projects	/project-manager/projects	PM, Admin recommended	Project list and progress.
Task	Tasks	/team-lead/tasks, /project-manager/tasks	PM, TL	Task assignment and tracking.
Team	My Team	role-wise /team	PM, TL	Assigned team member details.
Announcement	Announcements	role-wise /announcements	All roles	Admin/HR create; others view.
Support	Support Tickets	role-wise /support	All roles	Raise and track technical support issues.
Profile	My Profile	role-wise /profile	Employee, PM, TL and recommended all	Personal profile page.
Settings	Settings	/admin/settings	Admin, limited HR recommended	System and HR settings.
Assets	Asset Dashboard	recommended /assets	Admin, HR, view by others	Missing in current React project; should be added.
Assets	Manage Assets	recommended /assets/manage	Admin, HR	Asset list, add/edit asset, asset status.
Assets	Assignments	recommended /assets/assignments	Admin, HR	Assign assets to employees.
Assets	Replacements	recommended /assets/replacements	All raise, Admin/HR manage	Replacement/repair request tracking.

10. Database Model Requirements

Model	Important Fields	Purpose
User	id, username, email, password_hash, role, is_active, last_login	Authentication and role access.
Role	id, name, permissions	Role-based permission management.
Employee	id, user_id, employee_code, name, phone, department, designation, joining_date, manager_id, status	Employee master record.
Department	id, name, head_id, status	Company department list.
Designation	id, title, department_id, level	Employee designation list.
Attendance	id, employee_id, date, check_in, check_out, total_hours, status, remarks	Daily attendance records.
LeaveType	id, name, annual_quota, paid_status	Leave type configuration.
LeaveRequest	id, employee_id, leave_type, from_date, to_date, days, reason, status, recommended_by, approved_by	Leave application and approval flow.
Payroll	id, employee_id, month, year, basic, hra, allowance, bonus, tax, pf, other_deduction, net_salary, status	Monthly salary and payslip records.

Model	Important Fields	Purpose
Project	id, name, description, manager_id, start_date, end_date, status, progress	Project management.
Task	id, project_id, title, description, assigned_to, assigned_by, priority, due_date, status	Task assignment and tracking.
Asset	id, asset_code, category, brand, model, serial_no, purchase_date, status	Company asset master.
AssetAssignment	id, asset_id, employee_id, assigned_date, return_date, condition, status	Asset allocation to employees.
AssetRequest	id, employee_id, asset_id, request_type, description, screenshot, status	Asset replacement/repair request.
Announcement	id, title, body, category, priority, posted_by, posted_at, status	Company announcements.
SupportTicket	id, ticket_no, raised_by, title, description, screenshot, priority, status, assigned_to, resolution	Technical issue tracking.
Notification	id, user_id, title, message, read_status, created_at	User notifications.
AuditLog	id, user_id, action, module, record_id, timestamp	Track important changes.

11. Backend API Requirements

Module	Endpoint	Purpose
Auth	POST /api/auth/login/	Login user and return token/user role.
Auth	POST /api/auth/logout/	Logout user.
Users	GET/POST /api/users/	List and create users.
Users	GET/PUT/DELETE /api/users/{id}/	View, update or deactivate user.
Employees	GET/POST /api/employees/	Employee list and add employee.
Employees	GET/PUT /api/employees/{id}/	Employee detail and update.
Attendance	POST /api/attendance/check-in/	Mark check-in.
Attendance	POST /api/attendance/check-out/	Mark check-out.
Attendance	GET /api/attendance/	Role-based attendance records.
Leave	GET/POST /api/leaves/	List/apply leave.
Leave	POST /api/leaves/{id}/approve/	Approve/reject/recommend leave.
Payroll	GET/POST /api/payroll/	Payroll list and creation.
Payroll	GET /api/payroll/my-payslip/	Logged-in user payslip.
Projects	GET/POST /api/projects/	Project list and create project.
Tasks	GET/POST /api/tasks/	Task list and assign task.
Assets	GET/POST /api/assets/	Asset list and add asset.
Assets	POST /api/assets/assign/	Assign asset to employee.
Announcements	GET/POST /api/announcements/	List and create announcements.
Support	GET/POST /api/support-tickets/	List and raise support ticket.
Settings	GET/PUT /api/settings/	System settings.

12. Main Workflow Requirements

Workflow	Steps
Login Flow	User opens login page -> enters credentials -> backend verifies user -> backend returns role/token -> frontend saves session -> redirect to role dashboard -> sidebar shows role-specific menu.
Employee Creation Flow	Admin/HR adds employee -> employee record saved -> user account optionally created -> employee appears in employee list and reporting hierarchy.
Attendance Flow	Employee checks in -> attendance record created -> employee checks out -> total hours calculated -> HR/Admin/manager views reports.
Leave Flow	Employee applies leave -> TL/PM recommends if configured -> HR/Admin approves/rejects -> leave balance updates -> employee sees status.

Workflow	Steps
Payroll Flow	HR/Admin reviews attendance and salary -> payroll generated -> deductions and allowances calculated -> payslip published -> employee views own payslip.
Announcement Flow	Admin/HR creates announcement -> all roles receive announcement -> PM/TL/Employee can only view.
Support Ticket Flow	Any user raises issue with title, description and screenshot -> Admin/HR reviews -> status changes to In Progress/Resolved -> user sees ticket status.
Asset Flow	Admin/HR adds asset -> assigns to employee -> employee sees assigned asset -> employee raises replacement/repair request if needed -> HR/Admin resolves.

13. Frontend Requirements

- Use a clean, professional, consistent KIPL theme across all pages.
- Use fixed sidebar and top header layout for dashboard screens.
- All sidebar items must be route-linked and clickable.
- The sidebar menu must change based on logged-in role.
- All tables must include search, filter, status badge, and action buttons where required.
- All forms must include validation messages for required fields.
- Mobile view must be responsive with proper menu behavior and no overlap.
- Use reusable components for cards, tables, forms, badges, modals, and layouts.
- Replace dummyData.js with API data once backend is connected.
- Add missing Asset Management pages in the React project.

Recommended React Folder Structure

Folder	Purpose
src/components	Reusable UI components such as Button, Card, DataTable, Modal, FormField.
src/layouts	DashboardLayout, AuthLayout.
src/pages	Page-level components for each module.
src/routes	ProtectedRoute and route configuration.
src/services	API service files such as authApi, employeeApi, attendanceApi.
src/context	AuthContext and app-wide state.
src/utlis	Helpers for role permissions, date formatting, validation.
src/assets	Images, icons, logos.

14. Backend Requirements

- Create Django project with separate apps: accounts, employees, attendance, leave, payroll, projects, tasks, assets, announcements, support, settings.
- Use Django REST Framework serializers and viewsets for APIs.
- Implement role-based permissions on every API, not only frontend routes.
- Use PostgreSQL in settings.py and keep credentials in environment variables.
- Enable media upload for screenshots and employee/asset documents.
- Add proper validation in models and serializers.
- Add admin panel registration for all important models.
- Use migrations and seed initial roles: admin, hr, projectManager, teamLead, employee.

15. Security and Permission Rules

- A user cannot access a route or API outside their role permission.
- Sensitive data such as payroll should be visible only to Admin/HR and the owner employee.
- Passwords must be hashed and never stored as plain text.
- Every create/update/delete operation must be checked on backend permission level.
- Support ticket screenshots must be validated for file type and size.
- Use HTTPS in production.
- Use audit logs for important operations such as payroll changes, role changes, and employee deletion/deactivation.

16. Project Completion Analysis

Area	Current Status	Estimated Completion	Remarks
Frontend UI	Partially ready	55%	Main role pages exist, but many screens still need forms, API integration, validations, and polish.
Routing and Role Menus	Partially ready	65%	Protected routes and sidebar menus exist. Backend permission validation still required.
Backend	Not found in uploaded React zip	0%	Django backend must be created or connected.
Database	Not connected	0%	PostgreSQL models, migrations and APIs pending.
Assets Module	Missing/incomplete	10%	Required module should be added.
Support Module	Basic page exists	35%	Needs screenshot upload, status workflow and backend storage.
Overall Project	Partially complete	35-45%	Frontend foundation exists; backend/database and complete workflows pending.

17. Development Roadmap

Phase	Work	Output
Phase 1	Finalize requirement sheet, role access, page inventory and UI style guide.	Development clarity for all team members.
Phase 2	Clean React frontend structure, fix broken CSS, create missing pages, add reusable components.	Stable frontend.
Phase 3	Create Django backend apps and PostgreSQL models.	Backend database foundation.
Phase 4	Create REST APIs and connect React forms/tables with backend.	Real data saving and fetching.
Phase 5	Implement role permissions, authentication, validations and media upload.	Secure working HRMS.
Phase 6	Testing, bug fixing, responsive check and deployment.	Live project ready for demo/use.

18. Acceptance Criteria

- Every role logs in and lands on the correct dashboard.
- Every role sees only the allowed sidebar menu and pages.
- Unauthorized URL access redirects or shows access denied.
- Employee data saves in PostgreSQL and appears in employee list.
- Attendance records save correctly with check-in/check-out and reports.
- Leave requests move through apply, recommend, approve/reject and history flow.
- Payroll records are visible according to role permissions.
- Announcements can be posted only by Admin and HR.
- Support tickets can be raised by all roles and managed by Admin/HR.
- Asset management supports asset list, assignment, replacement and return flow.

- The UI is responsive and professional across desktop, tablet and mobile.

19. Final Notes for Team

This requirement sheet should be treated as the master reference for the HRMS Portal. Frontend developers should build pages according to this role-wise structure. Backend developers should create APIs and database models according to the modules defined above. No module should be considered complete until it is connected with backend APIs, stores data in PostgreSQL, and follows the role permission rules.